
EXECUTIVE TECHNOLOGY PORTFOLIO

SELECTED TECHNOLOGY PORTFOLIO

Enterprise technology, transformation and architecture at scale.

A curated executive portfolio spanning national procurement and supply-chain transformation, enterprise capability architecture, platform modernization, national inventory governance, enterprise integration, data and AI architecture, customer-service transformation, security and resilience, government workflow, education, digital health and 150+ major and mid-sized delivery engagements across Egypt and Oman.

18+ years

54 technology organization · 7 direct reports

EGP 150M annual IT budget accountability

150+ major & mid-sized Keyframe projects

FLAGSHIP PROGRAM · MOST RECENT

MedIQ

The defining enterprise program in my current career chapter: a national governmental procurement and supply-chain platform connecting procurement, supply, inventory, health entities, suppliers, enterprise integrations, governed workflows, data and executive visibility at national scale.

Enterprise architecture

Core modernization

E-Tender & workflow governance

Mobile electronic inventory

Dynamics 365 integration

Data, security & resilience

38K+

health-team
users

11K+

healthcare
facilities

2K+

suppliers

**~6
months**

end-to-end
core
replacement

Leadership & ownership: At UPA, Mahmoud leads a 54-person technology organization with seven direct reports and final technical/architecture decision ownership. Earlier-career labels distinguish work personally architected/implemented from substantial contribution within a delivery team. Client ministries and agencies are project contexts, not separate employers.

2026 portfolio refresh: new evidence now makes explicit the decision ownership behind enterprise capability architecture, inventory governance, the Dynamics 365 integration model, data/AI target architecture and customer-service operating-model transformation. Roadmap capabilities remain labeled as target/in-progress rather than presented as completed production functionality.

Current MedIQ evidence snapshot: 38,000+ users · 11,000+ healthcare facilities · 2,000+ suppliers. FY2024/25 evidence separately records 35,357 purchase orders, 938,242 recipient invoices and 89,761 warehouses/stores; period-specific metrics are kept distinct.

UPA

Supply chain / logistics

Logistics & Customs Workflow

End-to-end architecture, workflow and integration design spanning supplier shipments, supplier invoices, shipping-price assignment, release tracking, customs clearance, procurement and downstream warehouse operations; implementation delivered by the team under Mahmoud's leadership.

Enterprise workflow and supply-chain architecture ownership

UPA

Reporting platform

Centralized Telerik Reporting Architecture

Designed the centralized reusable reporting architecture and selected Telerik Reporting as the enterprise reporting technology, enabling configurable parameterized reports, standardized PDF/Excel output and 20+ deployed reports across application workflows.

Architecture design, technology selection and engineering leadership

UPA

Security & resilience

Security, Resilience & Production Operations

Defense-in-depth architecture with IAM/ RBAC, least privilege, segregation of duties, MFA/OTP, TLS/API controls, F5 BIG-IP, network segmentation, container security, monitoring/logging, PostgreSQL replication, backup/recovery and a DR environment.

Architecture and technical decision ownership

UPA

Data / AI

Data Platform, Analytics & AI Direction

Defined and owned the target enterprise data and AI architecture for MedIQ as a governed five-layer model: operational sources, validation/cleansing, analytical models, executive reporting/dashboards and phased AI. The roadmap covers demand forecasting, shortage and expiry risk, unusual-order detection, redistribution, supplier performance, anomaly detection and scenario simulation, with data quality and human review treated as prerequisites to scale.

Target data/AI architecture, governance and phased adoption ownership

UPA

Cross-government integration

Healthcare & Government Interoperability

Architected and governed cross-government integrations connecting MediQ with ECIC, MoH/HMIS, GAHAC and Blood Bank systems, defining integration design, data flows and technical governance; ECIC and Blood Bank are confirmed production integrations. Separately, personally designed and developed production healthcare REST APIs; also directed the completed EGP 11M Dynamics 365 integration.

Enterprise integration architecture and delivery-governance ownership

Flagship

UPA

Enterprise integration

Microsoft Dynamics 365 Enterprise Integration

Owned the integration study, architecture and decision model for the completed EGP 11M Microsoft Dynamics 365 program spanning master data, contracts and purchase orders, supply execution, goods receipt and inventory, finance and Power BI reporting. The integration approach used API-first patterns, middleware/integration services and governed database views, with mapping, reconciliation, security, SIT/UAT, rollback planning and controlled production transition.

Enterprise integration architecture, decision ownership and delivery governance

UPA

Operating model

Build vs Buy

Customer Experience & Technical Support Transformation

Owned the study and decision framework for a unified customer-service, complaints and technical-support model covering CRM, IVR/ call distribution and ticketing. Evaluated on-prem CRM versus specialist outsourcing and a phased hybrid model using data sovereignty, time-to-market, customization, CapEx/OpEx, resilience, scalability, knowledge retention and integration as decision criteria.

Decision ownership · build-vs-buy/TCO · target operating model

UPA

Enterprise architecture

Capability map

Enterprise Capability & Digital Transformation Architecture

Defined the target enterprise capability landscape linking supply, tendering, inventory, finance, assets, maintenance, e-pharmacy, complaints, content management and healthcare-facility services into a connected digital-platform direction, with governance, financial integration, mobile field work, interoperability, data warehousing and phased AI as cross-cutting transformation themes.

Enterprise capability architecture and transformation decision ownership

Flagship

UPA

Medical assets

Unified Assets & Maintenance

Digitized medical-device inventory and maintenance visibility linked to MediQ. Designed the MapTiler-based geographic asset-mapping architecture with QR/GPS location workflows and GS1 APIs for GTIN/GLN-based identification, and led the operating-model transition from paper to governed electronic maintenance. Evidence includes 242,816 devices across 164 entities plus structured adoption workshops and legal/process alignment.

Architecture, operating-model, technology selection and transformation ownership

UPA

National inventory

National Inventory Governance

Designed the national inventory-governance operating model across the wider MediQ warehouse landscape, defining strategic/central, main, regional and subordinate warehouse roles, request/supply controls and auditable intervention rules. A June 2026 baseline across three healthcare sectors covered 18,875 warehouses: 14,904 active (79%), 3,442 active without consumption (18.2%) and 535 inactive (2.8%), enabling targeted governance and remediation.

Operating-model, governance and decision ownership

Flagship

UPA

Mobile

Mobile MediQ / Electronic Inventory

Mobile-enabled field stocktaking and physical-to-system reconciliation with digital stocktake records. Operational evidence includes 20,513 app downloads, 17,236 workshop attendees and 13,226 inventory committees.

Platform, delivery and operational-governance leadership

Flagship

UPA

Workflow governance

MediQ E-Tender

End-to-end electronic tender/practice platform covering 10 linked steps, 30 operational stages, 15 Chairman approvals and 20+ governed roles/permissions, with segregation of duties, workflow gates, traceability and controlled award-to-framework-agreement outputs.

Enterprise workflow, procurement-governance and architecture leadership

Flagship

UPA

National scale

MedIQ Core Modernization

Full re-architecture and replacement of the national governmental procurement and supply-chain platform. MedIQ was announced as winner of the Pharma Awards 2026 - Digital Transformation Award. Modernized legacy solutions toward ReactJS/Node.js, C#/.NET APIs, PostgreSQL/Redis, Docker and analytics platforms while preserving service continuity; approximately three months core rebuild and about six months end-to-end.

Executive technology, architecture and transformation leadership · 100% in-house delivery

Integral

National scale

Oman National Educational Portal

National education platform with hands-on contribution across analysis, applications, databases, reporting, testing, deployment, rollout and support.

Senior hands-on developer and key technical contributor within the delivery team

Integral

Government workflow

Maktabi Electronic Correspondence

Large correspondence system covering creation, routing, forwarding, tracking, reporting, database components, testing, release and support.

Major hands-on developer within the delivery team

Integral

Award-winning

Recruitment System

Complete government recruitment solution from analysis and design through database, development, testing and production deployment.

Technical owner · solution recognized through the project award

Integral

Education

Student Incident Management

System for recording, following up and reporting student incidents across the education environment.

Technical owner

Integral

Education

International Schools Management

Requirements, application and database development, testing, deployment and support.

Core technical contributor

Integral

IT operations

IT Asset & Hardware Management

Hardware-focused inventory and operational control for the Ministry of Education environment.

Core technical contributor

Integral

Agriculture

Ministry of Agriculture Portal & Omani Dates Market

Portal plus purchasing and request workflows for the Ministry of Agriculture environment.

Technical owner

Integral

Membership & events

AI Wahat Club Management

Memberships and affiliates, facility reservations, approvals and event management for a Royal Court Affairs club assignment.

Core technical contributor · client project, not employment at the club

Keyframe

Product foundation

Reusable .NET CMS

Multilingual/RTL publishing, pages, articles, news, events, media galleries, permissions, workflows, search, SEO, analytics, resource files and integrations.

Sole application architect and developer

Keyframe

Health

Egypt Ministry of Health & Population Portal

Large portal with services, CMS, electronic library, FAQ, newsletter, polls, media, social integration, reporting, security and deployment.

Technical owner

Keyframe

Digital government

Egypt Ministry of Planning Portal

Analysis, SRS authorship, application architecture, database, CMS, development, testing and deployment within the Keyframe client engagement.

Technical owner and SRS author

Keyframe

Government suite

Egypt Government Digital Portals

Representative work includes MSAD, Rate Your Services, NMI, the national anti-corruption committee portal, Egypt Calendar, Elections in Egypt and surveys.

Technical owner · client assignments, not government employment

Keyframe

Conference platform

Arab Medical Union, UCHID, BGICC & Arab Organizers

Registration, attendance, sessions, speakers, research evaluation, certificates, communications, administration and reporting.

Sole application architect and developer

Keyframe

ITIDA award

Yashfeen Medical Portal

Full medical portal recognized through an ITIDA award during the Keyframe period.

Technical owner · award-recognized solution

Keyframe

Public safety

Traffic Reporting & Officers' Affairs

Complete solution for traffic reports and officers' affairs, from analysis and database design through testing and deployment.

Technical owner

Keyframe

Education

Electronic Examinations & Review

Electronic examinations and curriculum-review system for multiple educational certificate curricula.

Technical owner

Keyframe

Multi-country

Electrolux Websites

Reusable CMS customized for Electrolux in Egypt, Saudi Arabia, the UAE and Turkey.

Technical owner

Independent venture

Digital health

Salamat Healthcare Platform

Bilingual healthcare platform developed as an independent digital-health product venture during the Keyframe period.

Founder and sole product/application developer · separate from Keyframe client work

EXECUTIVE ARCHITECTURE SCOPE
Strategy, architecture and operating resilience.

Current leadership spans target-state architecture, technology roadmaps, build-vs-buy decisions, vendor governance, TCO, architecture risk, technical debt and value realization.

ARCHITECTURE

Decision Model

Requirements and constraints, alternatives, PoC when needed, technical decision, controlled implementation, testing and production readiness.

INTEGRATION & DATA

Enterprise Interoperability

REST APIs, contracts and schemas, mapping, authentication, bidirectional flows, reconciliation, exception handling, versioning and backward compatibility. Architecture ownership is kept distinct from team implementation; personally coded API work is claimed only for the separately confirmed healthcare REST API implementation.

TECHNICAL

ADVISORY / VENDOR / TOR

Government Technical & Commercial Governance

Led technical assessments and solution direction for government and healthcare stakeholders including EDA, Public Prosecution, Gomhouria Pharmaceuticals and MoH; owned solution scoping, architecture, sizing, vendor evaluation and technical acceptance, and directly prepared the UPA Microsoft AI/BI ToR.

SELECTED GOVERNANCE APPOINTMENTS

Leadership beyond platform delivery.

Formal national-level contributions complement the technology portfolio and demonstrate governance, policy and stakeholder depth.

2025-2026

Egypt Digital Transformation Committee

Egypt-side digital-transformation committee contribution supporting the World Bank Group B-READY Assessment (2025-2026), with technical input on digitization, modernization and national competitiveness.

2021-2024

National Blood Operations Oversight

Board-level national oversight contribution relating to digital systems, data integrity and technology standards in the blood-supply environment.

2021-2024

Pharmaceutical Track & Trace

National governance contribution supporting vision, compliance, interoperability, data integrity and technology-roadmap discussions across UPA, EDA and Ministry of Health stakeholders.

This portfolio prioritizes representative, defensible work and the projects that best demonstrate scale, architecture, transformation and delivery depth. Smaller engagements and confidential identifiers are deliberately omitted.